

Honey-LLM: Mentor Progress Report

A Self-Hardening Honeypot Defense Ecosystem for Conversational AI

Lifecycle Status: **85% Complete**

Phases 0–5 Verified Live | Phase 6 Active

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98.3% Jailbreak Detection In-the-wild benchmark	0.0% Benign False Positives 0 legitimate queries trapped	~2 ms Benign Sieve Latency Sub-ms Tier-1 fast-path	10.4 s Auto-Patch Time Capture to NeMo rule live
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1. Executive Overview & Problem Statement

Commercial LLM applications face severe vulnerabilities from multi-turn adversarial prompt injections, jailbreaks, and data-exfiltration probes. Traditional defenses rely on rigid, blunt-force blocking that alerts attackers and leaks perimeter logic. **Honey-LLM** resolves this through a three-pillared proactive defense: (1) a multi-tier intent sieve, (2) an isolated deceptive honeypot (*'Mirror Maze'*) that entangles attackers, and (3) an autonomous feedback loop that synthesizes permanent NeMo guardrails with zero downtime.

2. Verbal Presentation Script (for Reading Out in Front of Mentor)

Opening Pitch & Status:	"Honey-LLM is currently 85% complete with Phases 0 through 5 fully implemented and verified live. We are now executing Phase 6, our final red-teaming and empirical validation milestone."
Core Accomplishments:	• Two-Tier Intent Sieve: Resolves benign queries in ~2 ms via TF-IDF+LogReg while routing suspicious queries to an 8B Llama-Guard 3 custom policy (98.3% detection @ 0.0% FPR). • Isolated Deception Sandbox: Reroutes quarantined attackers to an LLM-driven decoy ('Sarah') in a zero-trust Docker container (passed 5/5 isolation smoke tests) leaking synthetic bait. • Autonomous Guardrail Synthesis: Automatically converts captured exploits into NVIDIA NeMo Colang rules and hot-patches the live gateway in 10.4 seconds with 0 downtime. • SOC Dashboard & Telemetry: Sub-second polling Next.js SOC interface tracking attack taxonomies, detection tiers, and measured attacker dwell time.
Remaining Scope (~15%):	"We are currently running automated PyRIT campaigns across 12+ obfuscation converters, executing container breakout audits, and compiling the empirical thesis evaluation figures."

3. Phase-by-Phase Implementation Status

Phase & Focus	Core Deliverables & Architecture	Status	Empirical Results & Exit Criteria
Phase 0 Project Setup	Monorepo scaffolding, FastAPI backend gateway, Next.js UI, Ollama client integration.	Completed	Verified end-to-end chat roundtrip; all health checks passing.
Phase 1 Threat Taxonomy & Infra	8-category threat taxonomy (`threat_taxonomy.md`); Apple Silicon M4 GPU memory validation.	Completed	Dual 8B models co-resident on 16 GB unified GPU; latency profile established.
Phase 2 Semantic Intent Sieve	Two-tier OR-ensemble (Tier-1 fast-path + Tier-2 Llama-Guard 3 8B custom injection policy).	Completed	98.3% in-the-wild detection, 0.0% benign FPR; ~2ms benign latency.
Phase 3 Mirror Maze Sandbox	LLM-driven 'Sarah' decoy, synthetic bait injection, sticky quarantine, zero-trust Docker sandbox.	Completed	Isolation smoke test 5/5 PASS (zero net escape; non-root uid 10001).
Phase 4 Guardrail Synthesis	Pattern distillation, NVIDIA NeMo Colang synthesis, semantic embedding matcher, hot-patching.	Completed	10.4 s patch time; zero-downtime hot reload; 0 FP regression.
Phase 5 Forensic SOC Telemetry	Dark SOC intelligence dashboard, dwell-time analytics, live event stream, token-gated admin panel.	Completed	<1s live poll refresh; verified 1m18s multi-turn dwell measurement.
Phase 6 Red-Teaming & Eval	PyRIT automated obfuscation campaigns, container penetration testing, load benchmarking.	In Progress (~15% left)	Target: Complete 12-converter evasion matrix & capstone documentation.

4. Detailed Breakdown of Remaining Work (Phase 6)

- **Step 6.1 — Automated Adversarial Stress Testing (PyRIT):** Executing 12+ prompt transformation algorithms (Base64, ROT13, leetspeak, Unicode confusables, character spacing) across 5 core attack categories to systematically chart evasion boundaries.
- **Step 6.2 — Sandbox Container Penetration Audit:** Executing automated privilege escalation and egress probing scripts from inside the Mirror Maze container to generate cryptographic/network proof of zero sandbox containment leakage.
- **Step 6.3 — End-to-End Latency & Concurrency Profiling:** Measuring throughput and response times under concurrent simulated traffic to prove negligible production overhead.
- **Step 6.4 — Final Capstone Thesis & Evaluation Report:** Consolidating benchmark confusion matrices, telemetry graphs, and defense-in-depth conclusions into the final submission.

5. Mentor Review & Feedback Notes

Mentor Comments / Action Items: